Predictive Forecasting of Care Load and Placement Demand in the Unaccompanied Alien Children (HHS UAC) Program

A Time-Series Machine Learning Framework for Early-Warning Capacity Management

Research Domain	Predictive Analytics, Time-Series Forecasting, Humanitarian Operations
Primary Data Context	U.S. Department of Health and Human Services — Unaccompanied Alien Children Program
Forecasting Horizon	Short-term operational forecasting with 7–14 day decision-support lead time
Models Evaluated	Random Forest Regressor, Gradient Boosting Regressor, statistical baseline
Primary Evaluation Metric	Mean Absolute Error (MAE)

Abstract

The Office of Refugee Resettlement (ORR) Unaccompanied Alien Children (UAC) program operates in an environment characterized by substantial operational volatility. Rapid changes in arrivals, transfers, discharges, and placement activity can place significant pressure on shelter capacity, case-management resources, and downstream placement operations. Conventional reporting approaches are primarily descriptive and reactive, limiting the amount of actionable lead time available to operational decision-makers. This study proposes a predictive intelligence framework for forecasting short-term care-load and placement demand using historical daily operational data and machine-learning regression techniques. The framework combines temporal feature engineering—including a one-day lag, a seven-day rolling mean, and a dynamic net-flow pressure indicator defined as transfers from U.S. Customs and Border Protection (CBP) minus discharges from HHS care—with ensemble regression models. A chronological holdout strategy is used to reduce the risk of temporal leakage and to approximate real-world future forecasting. In the reported evaluation, Random Forest achieved a mean absolute error (MAE) of 10.72 children, while an optimized Gradient Boosting configuration achieved an MAE of 9.85 children. Against a population scale exceeding 2,400 children, these results indicate low average absolute forecast error for short-term operational planning. The model is further operationalized through an interactive Streamlit decision-support application that enables scenario simulation, model comparison, KPI monitoring, and automated capacity-pressure diagnostics. The resulting framework is intended to support a transition from reactive capacity management toward earlier, evidence-based resource planning.

Keywords: Unaccompanied Alien Children, HHS, ORR, predictive analytics, time-series forecasting, machine learning, Random Forest, Gradient Boosting, capacity planning, early-warning system

1. Introduction

The management of unaccompanied children entering federal care requires continuous coordination across intake, transfer, shelter, case management, and placement functions. Operational conditions can change rapidly when the volume of children entering the care system increases faster than available discharge and placement capacity. Such conditions may produce short-term capacity constraints, increased reliance on emergency resources, and additional pressure on operational personnel.

A central challenge is the time gap between an operational change and the point at which its consequences become visible in conventional reporting. A descriptive dashboard can show that care loads have increased, but it may not provide sufficient lead time to prepare additional capacity. Predictive analytics offers an opportunity to supplement retrospective reporting with forward-looking estimates of near-term care demand.

This research therefore develops a structured forecasting pipeline that uses historical daily UAC operational metrics to estimate short-term changes in the population of children in HHS care. The study focuses on three questions: (1) whether recent care-load history provides meaningful predictive signal, (2) whether operational flow variables improve forecast performance, and (3) whether the resulting model can be translated into a practical early-warning decision-support system.

2. Research Objectives

1. Develop a reproducible data-engineering pipeline for historical daily UAC operational data.

2. Engineer temporal and flow-based predictors that capture persistence, trend, and system pressure.

3. Compare ensemble machine-learning regressors under a chronological out-of-sample validation design.

4. Evaluate model performance using Mean Absolute Error (MAE) and operationally interpretable error measures.

5. Translate model outputs into an interactive diagnostic application capable of supporting capacity-planning decisions.

6. Establish a foundation for future forecasting extensions incorporating additional exogenous indicators.

3. Data and Methodology

3.1 Dataset Overview and Data Hygiene

The analysis uses historical daily tracking information associated with the HHS UAC program. The preprocessing workflow is designed to preserve temporal integrity while reducing common data-quality issues. Records without valid timestamps are removed, population fields are converted to numeric representations, and comma-formatted values are sanitized before analysis. The resulting observations are sorted chronologically so that lagged variables and rolling statistics reflect the actual operational sequence.

3.2 Feature Engineering

Feature engineering is designed around three operational concepts: persistence, short-term trend, and net system pressure. The one-day lag captures the strong temporal dependence commonly present in daily population counts. The seven-day rolling mean reduces the influence of isolated daily fluctuations while preserving the underlying weekly trend. Finally, the net-pressure variable represents the balance between children transferred from CBP custody into HHS care and children discharged from HHS care.

Feature	Definition	Operational Interpretation
Lag_1	Care load at t−1	Immediate persistence of the care population
Rolling_Mean_7	7-day moving average of care load	Smoothed short-term population trend
Net_Pressure	Transfers from CBP − Discharges from HHS	Positive values indicate intake exceeds exits

Table 1. Core predictive features used in the forecasting framework.

3.3 Forecasting Architecture

The predictive task is formulated as a supervised regression problem in which engineered operational features are used to estimate the subsequent care-load level. Two ensemble learning approaches are evaluated. The Random Forest Regressor aggregates predictions across multiple decision trees and is well suited to nonlinear relationships and interaction effects. The Gradient Boosting Regressor builds an additive sequence of weak learners, with each subsequent learner emphasizing residual errors from earlier stages.

3.4 Temporal Validation Strategy

Because the observations form a time series, random train-test shuffling is inappropriate: it can expose the model to information from future periods during training. Instead, the final 60 chronological observations are reserved as an out-of-sample test horizon. This design more closely represents the operational scenario in which a model is trained on historical records and then used to forecast future conditions.

4. Model Evaluation

Model	Configuration	MAE (children)	Interpretation
Random Forest	100 estimators	10.72	Strong short-term predictive performance
Gradient Boosting	Optimized configuration	9.85	Lowest reported average absolute error

Table 2. Reported model performance on the chronological holdout set.

The Random Forest model produced an MAE of 10.72 children. Relative to a care population exceeding approximately 2,400 children in the reported evaluation context, this corresponds to an average absolute deviation of under 0.5% of the population scale. The optimized Gradient Boosting model achieved a lower MAE of 9.85 children, indicating a modest performance advantage under the reported configuration.

These results should be interpreted as short-horizon forecasting performance rather than evidence of universal model superiority. Model accuracy can change as arrival patterns, operational policies, reporting practices, and population dynamics change. Consequently, continued monitoring, rolling backtesting, and periodic retraining are recommended before production use.

5. Operational Deployment and Decision-Support System

To translate the forecasting workflow into an operationally accessible tool, the model is deployed through an interactive Streamlit application. The application is designed as a decision-support layer rather than a replacement for human operational judgment.

Component	Function

Dynamic simulation controls	Allows analysts to vary evaluation horizons and compare forecasting engines.
Model selection	Supports Random Forest, Gradient Boosting, and statistical baseline comparisons.
Executive KPIs	Displays analyzed-record counts, current care load, and forecast-error indicators.
Capacity stress diagnostic	Evaluates net pressure and highlights periods in which intake exceeds discharge activity.
Interactive visualization	Communicates historical and forecast trajectories for operational interpretation.

Table 3. Core components of the proposed decision-support application.

5.1 Early-Warning Logic

The capacity diagnostic uses net pressure as an interpretable operational signal. When transfers into HHS care exceed discharges, net pressure becomes positive, indicating that the active care population is under upward flow pressure. The application can therefore surface an early-warning condition before a sustained increase in the observed care population becomes fully visible in retrospective reports.

Importantly, a positive net-pressure signal should be treated as a screening indicator rather than an automatic declaration of crisis. Operational leaders should consider current capacity, geographic distribution, placement availability, staffing, and other contextual information before taking action.

6. Discussion

The findings support the premise that short-term UAC care-load dynamics contain substantial predictive structure. The effectiveness of the lagged care-load feature is consistent with population persistence: today's care population is strongly related to the recent population level. The seven-day moving average adds a smoother representation of trend, while net pressure provides a flow-based measure that can capture directional stress in the system.

From an operational perspective, the principal value of the framework is not simply a lower forecasting error. Its value lies in converting historical operational records into an interpretable signal that can be acted upon before capacity constraints become acute. This distinction is important in humanitarian and child-welfare environments, where decisions often involve lead times for staffing, transportation, facility coordination, and placement planning.

The results also highlight the importance of model governance. A predictive system used in a high-impact operational environment should include documented data lineage, validation procedures, monitoring for distribution shift, versioned model artifacts, and clear human-oversight procedures.

7. Limitations

1. The analysis is based primarily on historical operational variables and therefore may not capture external drivers of migration or placement demand.

2. A 60-observation holdout provides a useful out-of-sample test but does not by itself establish long-term stability across different operational regimes.

3. The reported MAE values depend on the selected feature set, model hyperparameters, data period, and preprocessing assumptions.

4. Net pressure is an informative flow indicator but does not directly measure available physical capacity, regional imbalance, staffing levels, or placement availability.

5. Forecasts should be used as decision-support outputs and not as deterministic predictions of individual child outcomes.

8. Future Research

Future iterations should extend the feature space beyond internal flow variables. Potential additions include regional-level indicators, policy changes, seasonal effects, weather variables, border-activity measures, facility-level capacity, placement pipeline status, and other exogenous factors that may influence demand. Hierarchical or regional forecasting could further improve resource planning by estimating not only the national care load but also where pressure is likely to emerge.

Methodologically, future work should evaluate walk-forward validation, probabilistic forecasting, prediction intervals, hyperparameter optimization, and model ensembles. Probabilistic forecasts would be particularly valuable because operational leaders need to understand not only the expected care load but also the range of plausible outcomes.

9. Conclusion

This study presents a practical predictive-analytics framework for short-term forecasting of care load and placement-related operational pressure in the HHS UAC program. By combining temporal features, flow-based pressure indicators, chronological validation, and ensemble regression, the framework demonstrates that machine learning can provide accurate and operationally interpretable short-term forecasts.

The reported results—MAE of 10.72 children for Random Forest and 9.85 children for the optimized Gradient Boosting model—indicate strong short-term predictive performance within the evaluated dataset and holdout horizon. More importantly, embedding the forecasting pipeline into an interactive Streamlit application transforms the model from an analytical experiment into a reusable decision-support capability.

The broader contribution of this work is the shift from retrospective monitoring toward anticipatory capacity management. With appropriate governance, continuous validation, and additional contextual variables, such a framework can help operational stakeholders identify emerging pressure earlier and make more informed resource-allocation decisions.

10. Technical Architecture Summary

Data Ingestion	Historical daily UAC operational records
Data Cleaning	Timestamp validation, numeric conversion, missing/corrupt record handling
Feature Engineering	Lag_1, Rolling_Mean_7, Net_Pressure
Validation	Chronological split; final 60 records as holdout horizon
Models	Random Forest Regressor; Gradient Boosting Regressor; baseline
Evaluation	Mean Absolute Error (MAE)
Deployment	Streamlit interactive diagnostic application
Decision Layer	Forecast visualization, KPI monitoring, net-pressure early warning

Appendix A. Reproducibility and Model Governance Recommendations

1. Maintain a versioned copy of the source dataset and document all preprocessing transformations.

2. Use walk-forward or rolling-origin validation for production model monitoring.

3. Track MAE and additional metrics such as RMSE, MAPE where appropriate, and forecast bias.

4. Monitor feature distributions and prediction errors for concept drift.

5. Retrain the model on a documented schedule or when monitoring thresholds indicate degradation.

6. Maintain an audit trail for model versions, parameter configurations, and deployment changes.

7. Use the system as decision support with qualified human review, especially when forecasts indicate potential capacity stress.

Research Note

The quantitative performance figures in this document (including MAE values and population scale) are based on the supplied study description. For formal academic or institutional publication, these values should be independently reproduced from the underlying dataset and code, and the final manuscript should include complete dataset provenance, experimental configuration, and bibliographic references.